



Customer Shopping Behavior Analysis

Uncovering insights from 3,900 purchases to drive strategic business decisions

Dataset at a Glance

3,900

Total Purchases

Transactions analyzed

18

Data Points

Features per customer

50

Locations

Geographic coverage

\$59.76

Avg Purchase

Mean transaction value



Data Preparation Journey

01

Data Loading

Imported dataset using pandas for analysis

02

Exploration

Used `df.info()` and `.describe()` for structure review

03

Missing Data

Imputed 37 missing review ratings with category medians

04

Feature Engineering

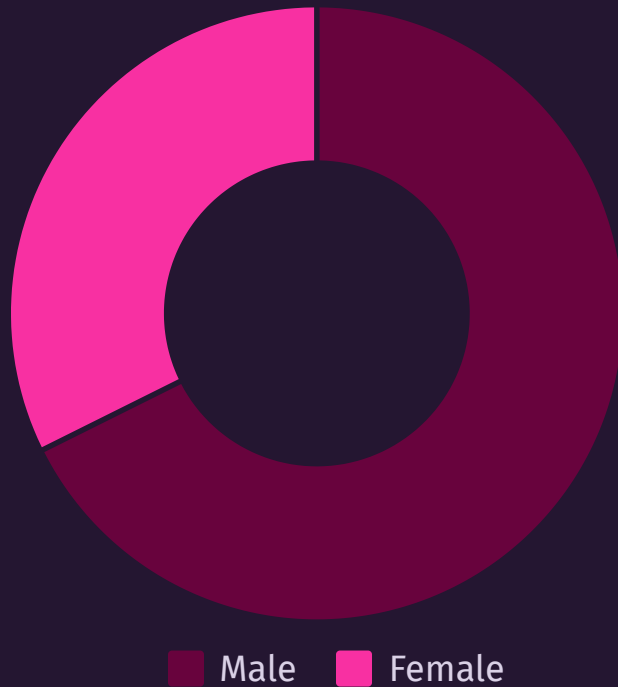
Created `age_group` and `purchase_frequency_days` columns

05

Database Integration

Connected to PostgreSQL for SQL analysis

Revenue by Gender



Male Customers Drive Revenue

Male shoppers generate 68% of total revenue, more than double female contribution

\$157,890 from male customers vs **\$75,191** from female customers

Top-Rated Products



Gloves

3.86 rating



Sandals

3.84 rating



Boots

3.82 rating



Hat

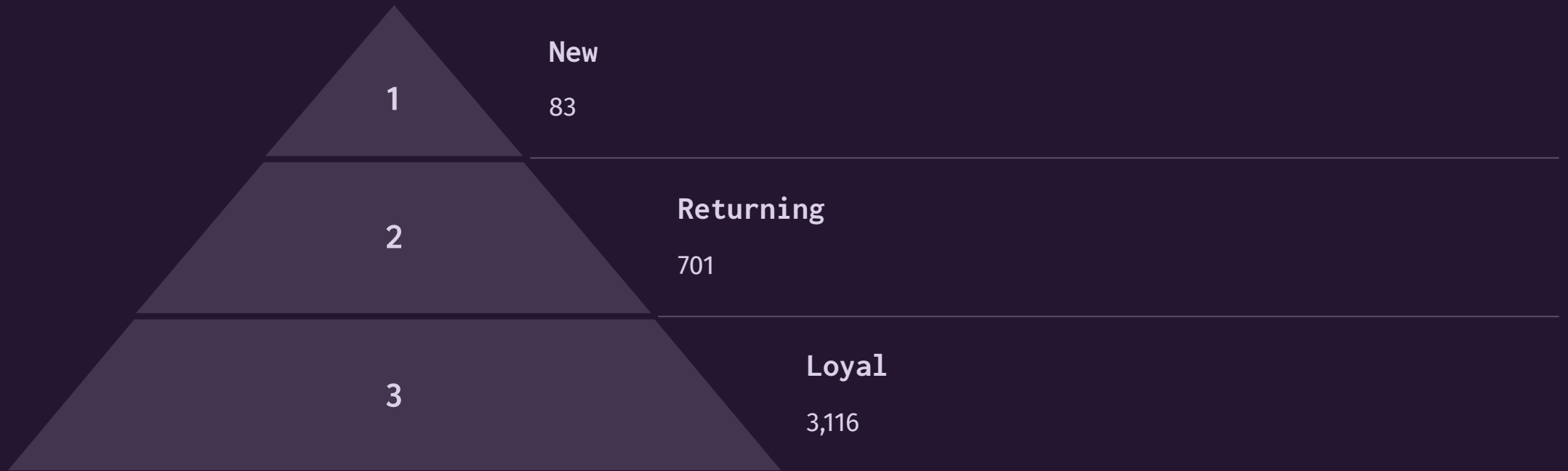
3.80 rating



Skirt

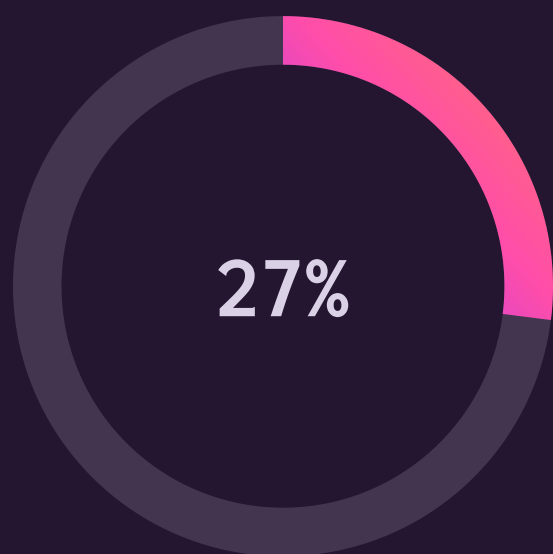
3.78 rating

Customer Segmentation Insights



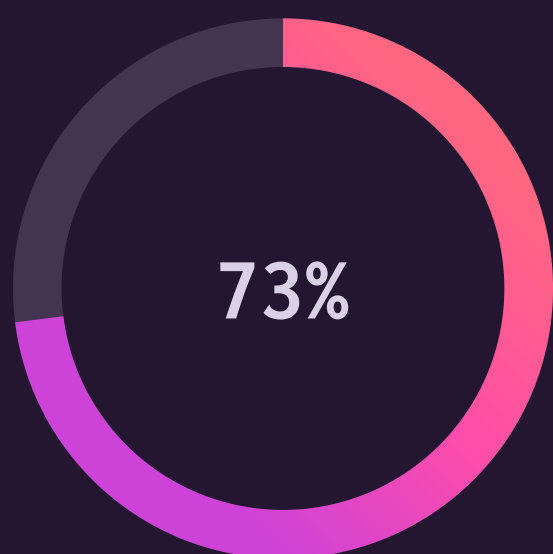
80% of customers classified as **Loyal** based on purchase history — a strong foundation for retention strategies

Subscription vs Non-Subscription



Subscribers

1,053 customers



Non-Subscribers

2,847 customers

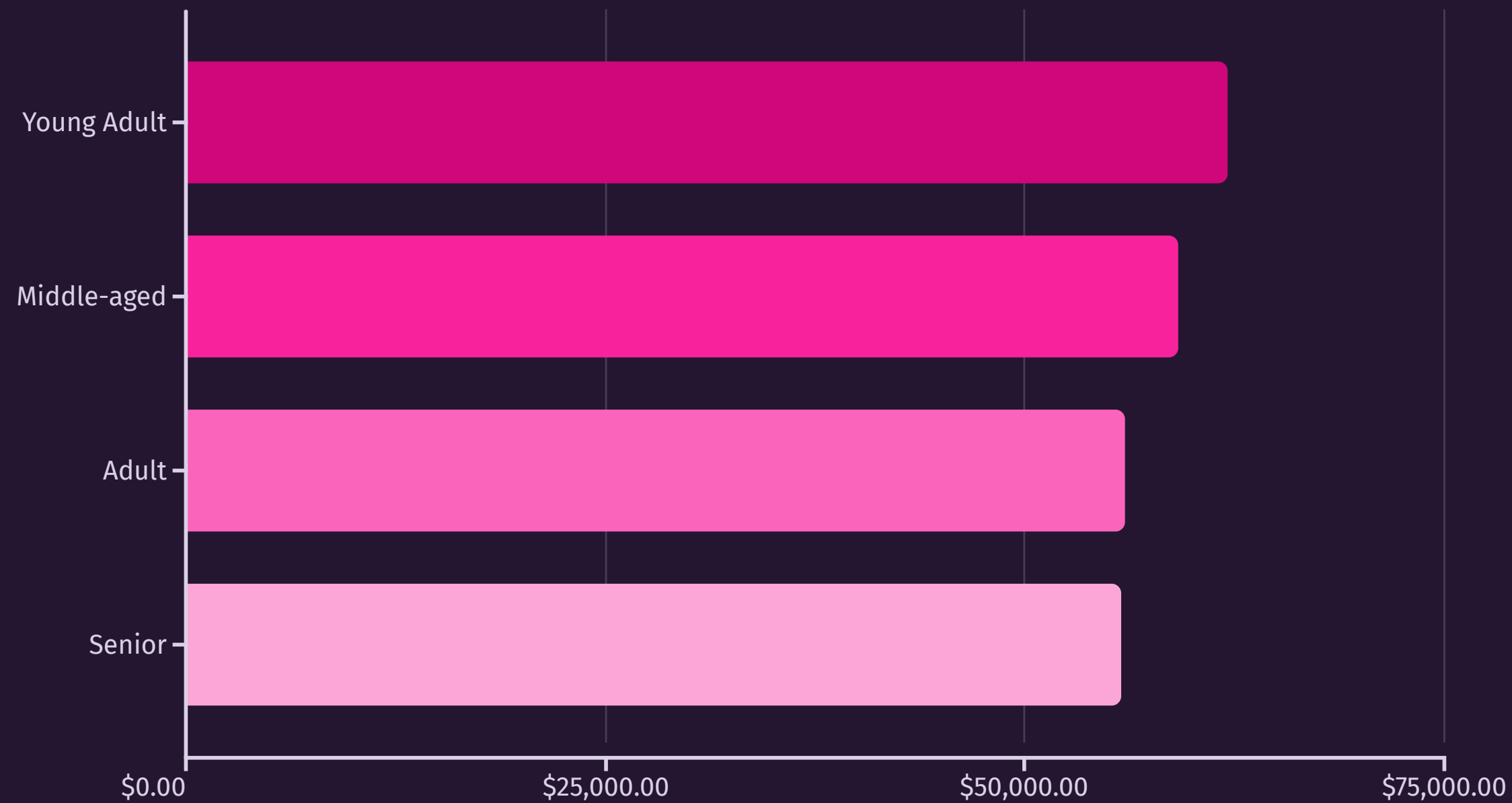
Similar Spending Patterns

Subscribers: **\$59.49** avg spend

Non-subscribers: **\$59.87** avg spend

Despite similar spending, non-subscribers generate **2.7x more total revenue** due to volume

Revenue by Age Group



Young adults lead revenue generation at **\$62,143**, with relatively balanced contribution across all age segments



Discount Strategy Analysis

High-Value Discount Users

839 customers used discounts but spent **above average**

Discounts don't always mean lower revenue

Discount-Dependent Products

Top 5: Hat (50%), Sneakers (49.66%), Coat (49.07%)

Nearly half of these purchases require discounts

Shipping Impact

Express shipping: **\$60.48** avg purchase

Standard shipping: **\$58.46** avg purchase

Strategic Recommendations



Boost Subscriptions

Promote exclusive benefits to convert 73% non-subscribers into recurring revenue



Loyalty Programs

Reward repeat buyers to strengthen the 80% loyal customer base



Review Discount Policy

Balance sales boosts with margin control for discount-dependent products



Targeted Marketing

Focus on high-revenue young adults and express-shipping users